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Introduction

The venerable FRG-8800 is now over 30 years old and inevitably things start to go wrong. So it was with my '8800 when I found the squelch wasn't working and worst of all it had failed so there was no audio. I could jiggle the pot and sometimes get it to work but it was very touchy and cleaning solvent made no difference. Checking the manual shows that the squelch is a dual gang 10K LIN pot. The pot is a strange physical construction and is PCB mounted with the pins for both pots in a single row. I've never seen anything like it. How hard could it be to find a replacement K162Y0Z01-10KBx2 pot? I'm sure you already know the answer.

The quick fix

I thought that perhaps I might simply sacrifice the squelch function by shorting the appropriate pins on the pot. Whilst reasonable for AM/SSB it's probably not suitable for FM as there can be long periods of inactivity between transmissions and the hiss would get annoying.

Indeed, this is a very simple repair only requiring the removal of the bottom cover and bridging the two centre pins of VR01 on the back of the PUSH SW UNIT. This opens the AM squelch. Bridging the two upper pins as seen in the image at left would open the FM squelch.

Of course this mean the radio no longer behaves as designed.

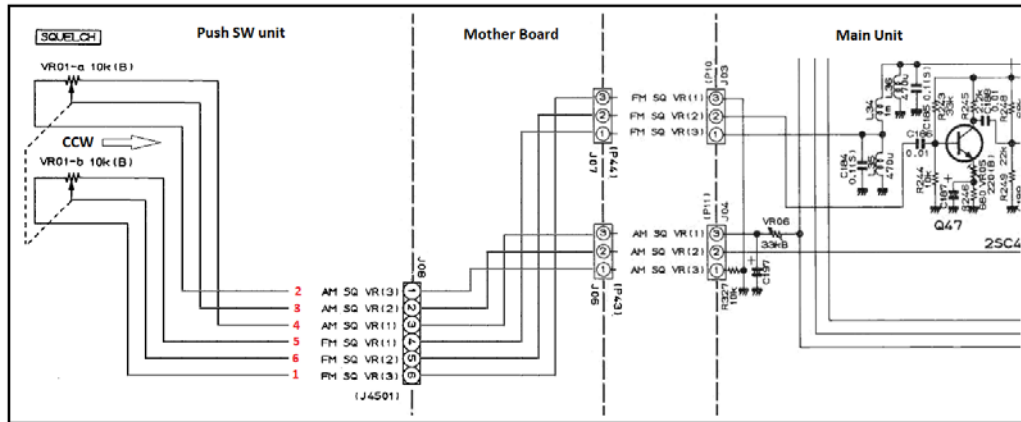
Read on if you want to see my approach to replacing the squelch pot.



Disclaimer

This project is offered as is with no guarantees and with the understanding that it may contain errors. Should you decide to do this to your radio I assume that you understand the technical details and are capable of carrying out this work in a competent fashion. Should you break your radio, remember who took the covers off and started this activity. I will not be held responsible for you breaking your radio. Do not start this project unless you are sure you are capable of doing the work

Below is a summary of the squelch circuit wiring. It will come in handy later on. The red numbers correspond to those in the other image on the page. Also note the direction indicator for the pot.



Shorting pins 3 & 4 will open the AM squelch, shorting pins 5 & 6 will open the FM squelch. If that's enough for you then stop here, your work is done.

Replacement

Have no doubt that this is a big job as you have to pull the front panel off the radio then pull boards off the front panel. It's probably a three hour job if everything goes well.

You need to get the complete PUSH SW UNIT out of the radio to replace the squelch pot. *Skip forward to the disassembly section if you have never dismantled your radio then return here when you have the PUSH SW UNIT in your hands.*

Remove the metal bracket supporting the three pots then unsolder the old AF pot and install the new one.

Take special note that the pot connections on the PCB are not straight forward and since the circuits are spread over a few pages it's hard to follow what is going on. The most important task is to figure out how to wire the replacement pot to the PUSH SW UNIT.



Note how the solder pads for the pot are connected to J08 on the mother board. The summary circuit above will help you work out the wiring for the replacement pot.

The replacement pot



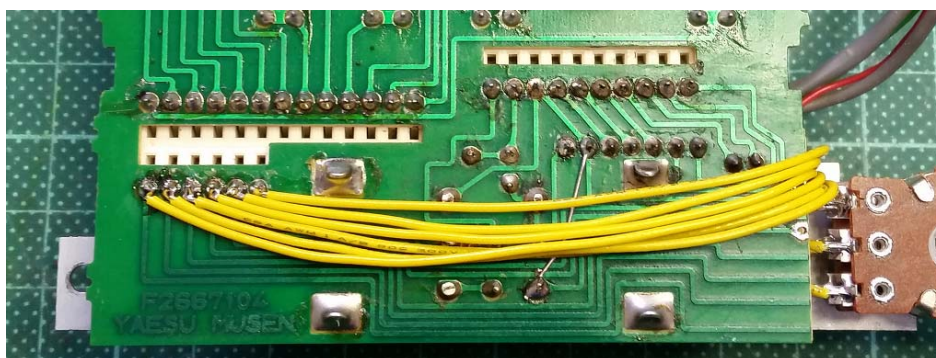
I wanted to maintain the functions and outer cosmetic appearance of the radio but accepted that there may have to be some changes made inside to accommodate the replacement.

I was lucky enough to find an electrical equivalent on eBay from a seller called *mc_manager* in the UK. It was a different shape from the original but at least it had two gangs and I'm sure I could make it fit.

The replacement pot body was larger and deeper than the original pot which means it would not fit under the PUSH SW UNIT like the original. I decided that I would cut away a section of the PCB and use wires to connect to the remainder of the PCB.



I cut the board by scoring it with a hobby knife then snapping off the unwanted section. Note that I moved the small jump to be sure it wasn't affected. The next step was to wire the pot directly to J08 with some short wire.



Taking the time to trace out the squelch summary circuit helped with this task and made it really easy to work out which wire goes where. I could have used some tape or hot glue to keep the wires in place but once the front panel was assembled they were quite secure and I didn't see the point.

Assemble your radio and enjoy your handiwork.

Disassembly

Remove both covers and the four screws that hold the front panel to the chassis then lean the panel forward carefully. Remove the plugs from the MOTHER BOARD UNIT that tether the front panel to the chassis. As you pull each plug, check the number on the side and write it next to each socket on the board. Although the connections are recorded in the service manual doing it this way makes it easier when you put the radio back together.

Once you are done put the chassis to one side and concentrate on the front panel.

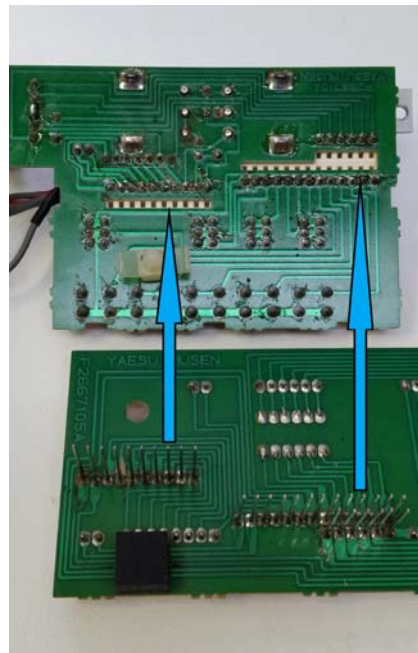
Remove the AF, Tone and squelch knobs only. Disconnect the cable to the POWER SW UNIT and “carefully” disconnect the speaker along with the VFO cable and the MEM switch cable.

The MOTHER BOARD UNIT must be unplugged and removed from the KEY SW UNIT and the PUSH SW UNIT.

There are two PCB standoffs holding the motherboard to the other two boards and a number of pins that plug into sockets on the lower boards. They are a REALLY tight fit but they will come out. Release the standoffs and ease the mother board up. Use a piece of wood dowel or something that won't damage the tracks to carefully lever the mother board up from the two boards. Start at the KEY SW end to get an idea of the force required. GO SLOWLY! With too much leverage you could break the boards or bend the pins. With

care and patience you will get the mother board out.

Next, remove two screws securing the metal plate holding the squelch pot to the front panel. Look carefully and you will see a number of plastic clips holding the PUSH SW UNIT to the front panel. Ease these clips back gently as you lift the board up. Work slowly and carefully to protect the clips from breakage. Take care as once you remove the board, the buttons beneath can easily fall out and be lost. Now return to page 2 and continue with the replacement.



How the mother board fits to the PUSH SW UNIT